Exp-16:

Sobel algorithm using Open CV to filter the input image.

Code:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
image = cv2.imread('/content/cvexp.jpg', 0)
sobel_x = cv2.Sobel(image, cv2.CV_64F, 1, 0, ksize=3)
sobel_y = cv2.Sobel(image, cv2.CV_64F, 0, 1, ksize=3)
sobel_magnitude = cv2.magnitude(sobel_x, sobel_y)
sobel_magnitude_display = np.uint8(sobel_magnitude)
sobel_x_display = np.uint8(np.absolute(sobel_x))
sobel_y_display = np.uint8(np.absolute(sobel_y))
plt.figure(figsize=(20, 5))
plt.subplot(1, 4, 1)
plt.imshow(image, cmap='gray')
plt.title('Original Image')
plt.axis('off')
plt.subplot(1, 4, 2)
plt.imshow(sobel_x_display, cmap='gray')
plt.title('Sobel X')
plt.axis('off')
plt.subplot(1, 4, 3)
plt.imshow(sobel_y_display, cmap='gray')
plt.title('Sobel Y')
plt.axis('off')
plt.subplot(1, 4, 4)
plt.imshow(sobel_magnitude_display, cmap='gray')
plt.title('Sobel Magnitude')
plt.axis('off')
plt.tight_layout()
plt.show()
```

Output: